

Licensed Electrician's Theory (LET)

Sample Paper (2021)



Candidate Surname	
Candidate Given Names	

Reference Material

- AS/NZS 3000:2018 Wiring Rules
- AS/NZS 3012:2019 Electrical installations – Demolition and Construction sites
- AS/NZS 3008.1.1:2017 Electrical installations – Selection of cables
- Electricity Safety (General) Regulations 2019
- AS/NZS 4836:2011 Safe working on or near low voltage electrical installations and equipment

Instructions

- Personal notepads and paper are not permitted.
- Pens only must be used. Answers in pencil may not be marked.
- Do not remove any sheets from this assessment paper or the room.
- Papers with no name or signature will not be marked.
- Units must be shown to obtain full marks.

Results

Candidates need to obtain 75% or more to pass this assessment. If a mark of 74% or less is achieved, a minimum of 14 days is required before you are permitted to re attempt the assessment.

I, the above named candidate confirm:

- I understand the instructions provided to me.
- do not have any unauthorised materials in my possession
- have not attempted the Licensed Electrician's Theory Assessment at any venue within the past 14 days.

Candidate			
	Print name	Signature	Date

Working Time: 2 hours and 15 minutes

At the end of this time you will be asked to stop.

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	Total
Mark																			

Final Percentage	Pass/Fail

Supervisor			
	Print name	Signature	Date
Assessor			
	Print name	Signature	Date
Reviewed by (If necessary)			
	Print name	Signature	Date

AS/NZS 3000 WIRING RULES

In the following **four** Wiring Rules questions:

- You are required to write the Wiring Rules Clause and/or Table number in the space provided.
- The correct Wiring Rules Clause and Subclause must be given, e.g. 3.5.2(b)(i).

The correct answer to both parts must be given to obtain full marks.

Question 1. Every water heater that is fixed wired shall be provided with an independent isolating switch. Where is this isolation switch to be installed?

Wiring Rules Clause Number: _____ [2 + 2 = 4 Marks]

Question 2. Electrical equipment that forms part of an installation shall be selected and installed to operate in a safe and reliable manner under what conditions?

Wiring Rules Clause Number: _____ [2 + 2 = 4 Marks]

Question 3. Wiring enclosures installed on roofing material shall not be installed in a manner that promotes the accumulation of debris. List one way this may be achieved?

Wiring Rules Clause Number: _____ [2 + 2 = 4 Marks]

Question 4. A lamp holder on a flexible pendant is installed in a refrigeration room. What is the minimum distance it can be suspended from the floor or ground?

Wiring Rules Clause Number: _____ [2 + 2 = 4 Marks]

AS/NZS 3012 CONSTRUCTION AND DEMOLITION SITES

In the following **two** AS/NZS 3012 questions:

- You are required to write the Standard's Clause number and/or Table number in the space provided.
- The correct Clause and Sub-clause number must be given. e.g. 2.10.2 (f).

The correct answer to both parts must be given to obtain full marks.

Question 5. . What is the minimum lux level required in a walkway on a construction site?

Standard Clause Number: _____ [2 + 2 = 4 Marks]

Question 6. What must a single phase socket outlet in a portable socket outlet assembly be individually controlled by?

Standard Clause Number: _____ [2 + 2 = 4 Marks]

ELECTRICITY SAFETY (GENERAL) REGULATIONS 2019

In the following Regulation question, you are required to:

- write your answers on the line/s below each question
- write the complete Regulation and Sub-Regulation number, if applicable, in the space provided, e.g. 401(e)(3).

The correct answer to both parts must be given to obtain full marks.

Question 7. A person who employs an apprentice must ensure any electrical installation work carried out by the apprentice is carried out under effective supervision in accordance with what?

Regulation Number: _____ [2 + 2 = 4 Marks]

ELECTRIC SHOCK SURVIVAL

Question 8. You are applying first aid to a person who has received an electric shock. The power is off, and the emergency services have been called. The victim is unconscious, and you want to check their airway is clear. How do you do this?

1. _____
2. _____

[2 + 2 = 4 Marks]

CABLE SELECTION**Question 9.**

THREE four core and earth X90 insulated and sheathed copper cables, are connected in parallel to supply a three-phase distribution board with a total maximum demand of 270A.

The cables are buried direct in the ground at a depth of 0.6m and are spaced 0.45m apart. They are protected by a circuit breaker.

- (i) Neglecting voltage drop, what is the minimum cable size which can be installed for this circuit.
- (ii) If the cables were spaced 0.15m apart instead of .45m, what is the minimum cable size which can be installed for this circuit?

All calculations including the final answer must be completed to a maximum of **two decimal places**.

Table details and units must be shown below to obtain full marks. Part (ii) is over the page.

Part (i)

	Answer		Answer		
Table 3 (____?)		Item			
Table		Column			Answer
Derate/rating table		Column		Factor	
Derate/rating table		Column		Factor	

Part (i) Answer:

Question 9. Part (ii)

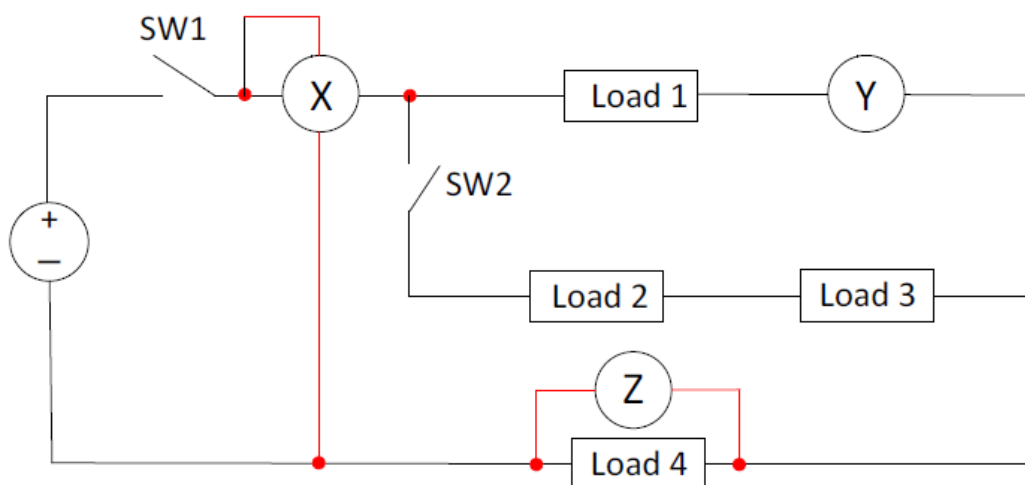
	Answer		Answer		Answer
Derate/rating table		Column		Factor	

Part (ii) Answer:

[1 + 2 + 1 + 1 + 2 + 1 = 8 Marks]

DC CIRCUITS

Question 10.



The following values apply to the diagram above:

The supply current is 800mA

Voltage across Load 1 is 96V, Current through Load 3 is 320mA, Load 4 is 80Ω

Using these measurements, calculate the meter readings for the following when the switches SW1 and SW2 is closed:

- (i) meter X,
- (ii) meter Y
- (iii) meter Z

All calculations including the final answer must be completed to a maximum of **two decimal places**.

Calculations:

(i) Meter X:

(ii) Meter Y:

(iii) Meter Z:

[2 + 2 + 2 = 6 Marks]

MAXIMUM DEMAND**Question 11.**

Calculate the Maximum Demand for the mains of a single phase 230V domestic residence.

The load connected to the switchboard is:

- 2 circuits of 8 - 10A double socket outlets
- 2 single 10A socket outlets
- 1 16A charger for an electric vehicle
- 1 5kw storage water heater
- 2 circuits of 18 – LED lighting points
- 1 – 3.3kW oven

All calculations including the final answer must be completed to a maximum of **two decimal places**.

All relevant table details, including table, column and load groups used.

Calculations and units must be shown to obtain full marks.

Table		Column	
Equipment	Load Group	Calculation	Maximum Demand
Total Maximum Demand:			

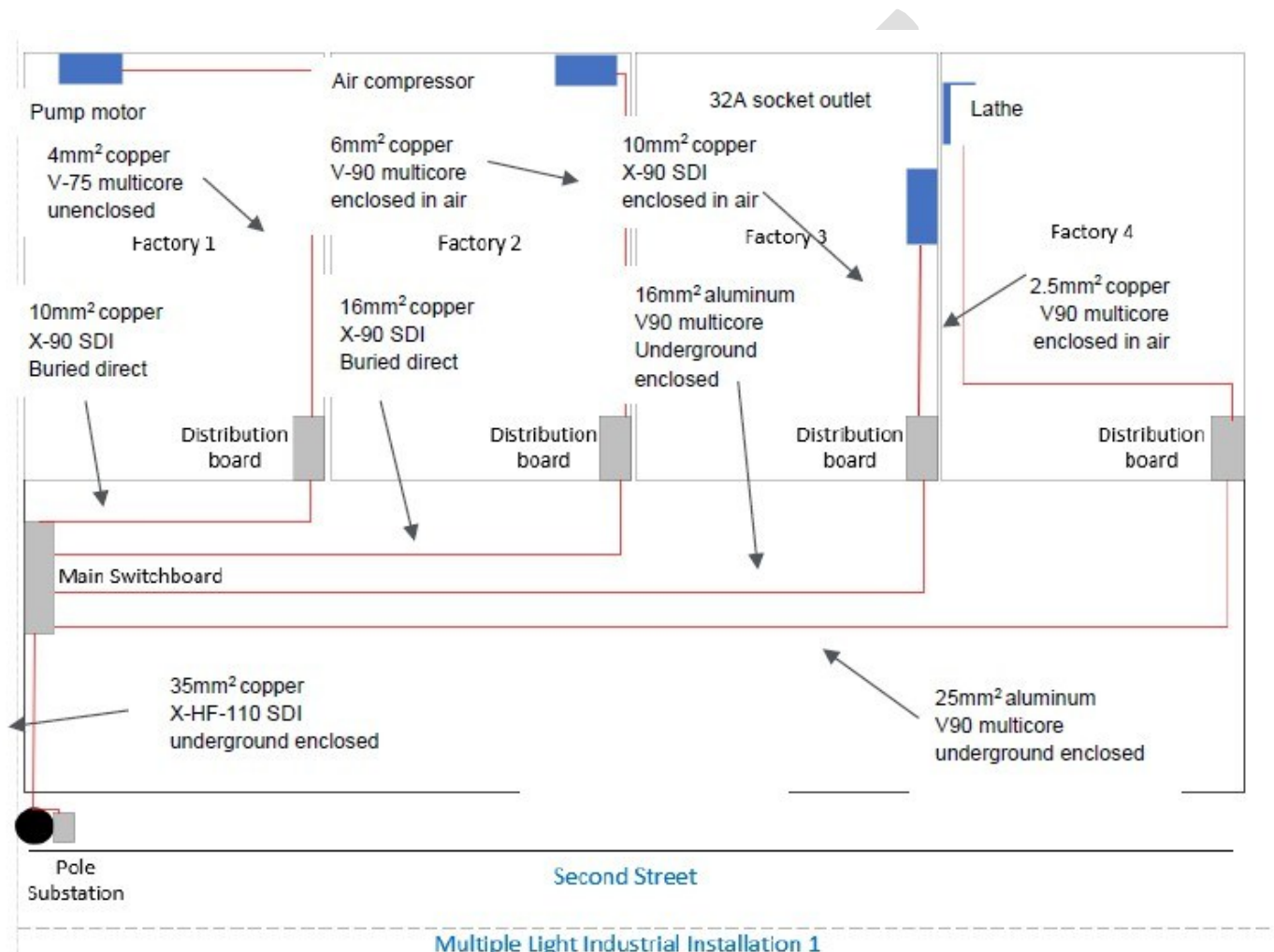
[1 + 1 + 1 + 1 + 1 + 1 + 2 + 1 = 8 Marks]

VOLTAGE DROP

Question 12.

In a 400/230V, three-phase factory installation, a three-phase 32A socket outlet is supplied from a sub-circuit originating at a distribution board in factory 3. The socket outlet is installed 15 meters away from the distribution board.

A site plan of the three-phase domestic installation is shown below, all the circuits are operating under normal conditions.



Client: Darren Wall	Location	Cable	Distance	Maximum Demand	Voltage
Drawing title: Site plan	Main Switchboard	Consumer Mains	16m	128A	400/230V
Date: 5 November 2024	Factory 1 Distribution board	Sub Mains	14m	57A	400/230V
Location: 65 Imagine St, Carlton	Factory 2 Distribution board	Sub Mains	22m	48A	400/230V
	Factory 3 Distribution board	Sub Mains	45m	62A	400/230V
	Factory 4 Distribution board	Sub Mains	50m	55A	400/230V

Using the information provided above, calculate the total voltage drop from the point of supply to the socket outlet terminals.

All calculations including the final answer must be completed to a maximum of **two decimal places**.

All relevant table details, calculations and units must be shown to obtain full marks.

Cable	Table	Column	Vc	Calculation	Vd
Consumer Mains					
Sub-mains					
Final Sub-circuit					

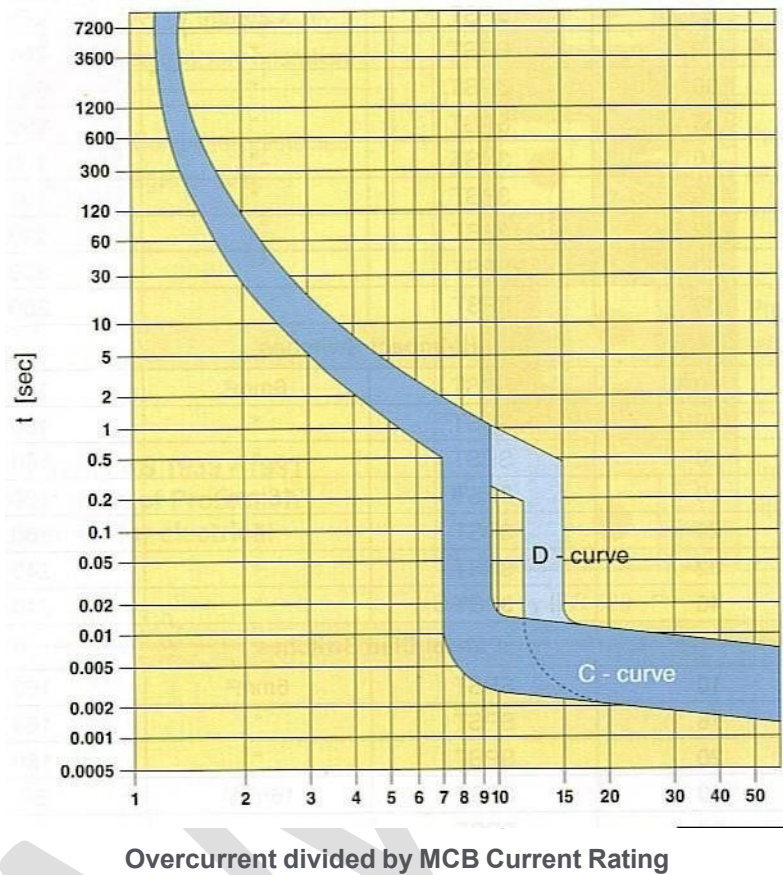
Answer Total Vd:	
------------------	--

[1+1+1+1+1+1+1+1+1+1 = 10 Marks]

OVERLOAD AND SHORT CIRCUIT CALCULATIONS

Question 13.

What are the minimum and maximum tripping times for a 25A Type C miniature over-current circuit breaker which is subjected to an over-current of 100A?



Overcurrent divided by MCB current rating:	
Minimum time:	Maximum time:

[1 + 1 + 1 = 3 Marks]

OVERLOAD AND SHORT CIRCUIT CALCULATIONS**Question 14.**

The main switchboard of a 400/230V industrial installation is directly supplied from a 500KVA transformer which has a prospective fault current of 15,560A per phase.

Submains supply a distribution board from the main switchboard.

The following information is known:

Impedance of the Consumer Mains = 0.0052Ω

Impedance of the Sub-mains cables = 0.028Ω

Determine the prospective fault current at:

- (i) the main switchboard; and
- (ii) the distribution board.

Work impedances to 5 decimal places.

All calculations must be shown to obtain full marks.

Transformer Impedance:	
MainSw/Bd:	
Dist/Bd:	

[(2+1) + (2+1) + (2+1) = 9 Marks]

RESIDUAL CURRENT DEVICES**Question 15.**

A 30mA Residual Current Device is to be installed to protect two (2) circuits of 10A socket outlets and a 15A socket outlet. The maximum demand of the three circuits is 35A. Each circuit is protected by a 16A circuit breaker.

State the minimum current rating of the Residual Current Device.

Wiring Rules Clause Number: _____

[1 + 2 = 3 Marks]

MOTORS AND STARTERS**Question 16.**

CIRCLE the letter in front of the statement you consider to be the most correct.

The following motor starters limit the current of the motor on initial startup, with the exception of:

- A. star/delta
- B. Auto-transformer
- C. primary resistance
- D. direct on line (DOL)

[2 Marks]

AS/NZS 4836:2011**Question 17.**

This question relates to AS/NZS 4836:2011.

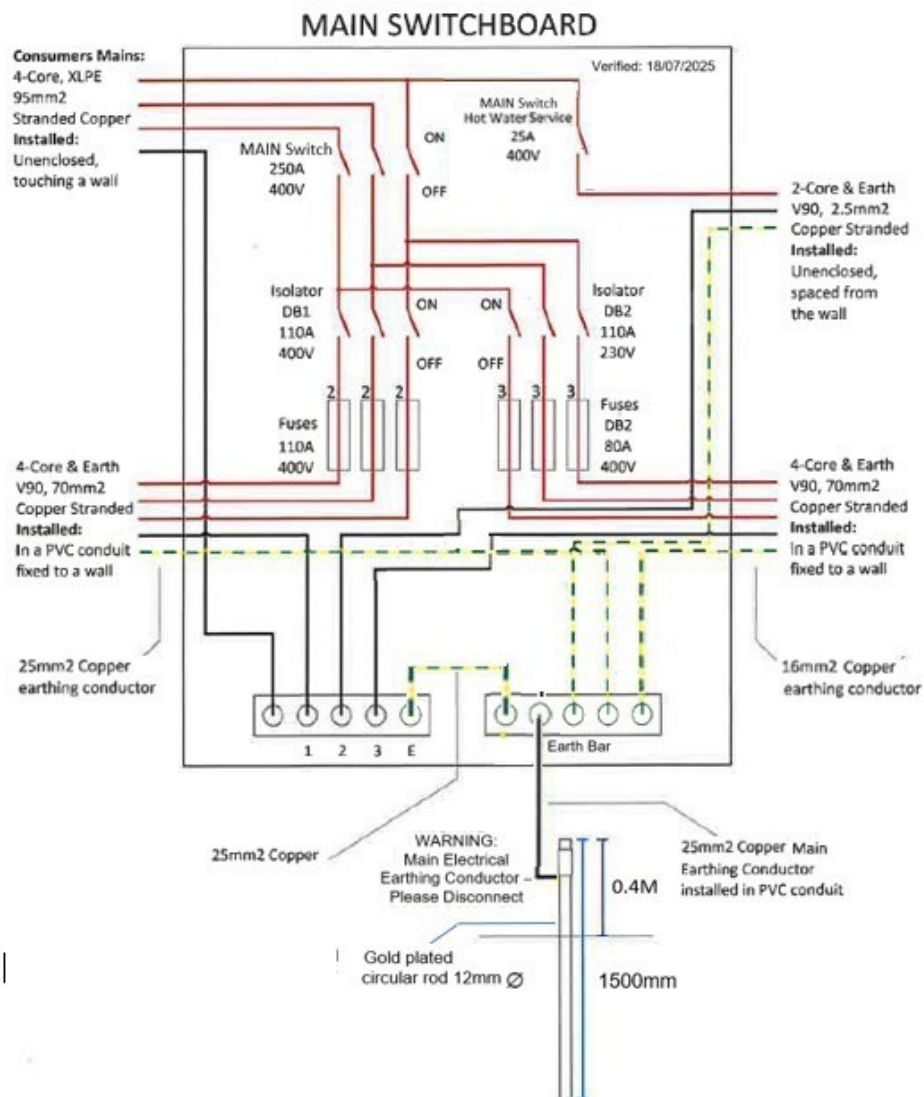
When isolating a supply and the risk assessment identifies the need for bonding, the conductors shall be bonded together and connected to where?

Standard Clause Number: _____

[2 + 2 = 4 Marks]

INSTALLATION DEFECTS - NON DOMESTIC

Question 18.



The MAIN SWITCHBOARD is supplied directly from a pole-mounted transformer via consumers mains installed unenclosed on a cable tray fixed directly to a wall and enters directly into the switchboard enclosure.

It supplies two distribution boards, each having a connected load with a calculated maximum demand of 100A per phase, and a single-phase water heater rated at 25A. the water heater has been deemed not at risk of electric shock.

All cables are installed as described on the drawing.

Assume any MIMS cables are earthed in accordance with the Wiring Rules and are capable of maintaining supply to the equipment when exposed to fire and mechanical damage.

All fuses are of HRC type.

All screws in bars & links are 60% of the tunnel terminal diameter.

A metal portion separates all switchgear from each other.

Complete the table on the following page.

Question 18. Continued.

Use the diagram on the previous page.

List **FIVE different defects** together with the contravened Wiring Rules Clause/Table number in the table provided below.

Note: Only the first five defects will be considered.

DEFECT DETAILS	WIRING RULE CLAUSE/TABLE No.

[5 x (2 + 1) = 15 Marks]

APPENDIX L
ELECTRIC SHOCK SURVIVAL—Australia

A

EMERGENCY !—Electric shock may cause cardiac arrest.

DANGERS

Check for your own safety and the safety of the casualty and bystanders.

HIGH VOLTAGE—Wait until the power is turned off.

LOW VOLTAGE—Immediately switch off the power. If this is not practicable, pull or push the casualty clear of the electrical contact using material, such as wood, rope, clothing, plastic or rubber. Do not use metal or anything moist.

RESPONSIVENESS

Check for response (verbal and tactile stimuli), touch and talk.

SEND/Shout FOR HELP

Send a bystander to DIAL 000 Ambulance

If available send for Automatic External Defibrillator (AED)

If alone shout for help.

AIRWAY Place the casualty on his/her back.

Tilt the head back and raise the chin forward.

BREATHING Check for normal breathing, observe chest movement, listen and feel for breathing.

Give two initial breaths.

In the absence of normal breathing, if no one has gone for help, place casualty in recovery position and go for help.

CIRCULATION

Position hands on centre of the chest.

Give 30 chest compressions followed by 2 breaths. Depress breastbone 1/3 the chest depth (approx 4 cm or 5 cm) at the rate of 100 compressions a minute.

As soon as available attach AED and follow its instructions.

Continue CPR, 30 compressions: 2 breaths.

When casualty's normal breathing returns cease resuscitation and move the casualty into the **recovery** or **coma** position.

Keep a constant watch on the casualty, to ensure that they do not stop breathing again, until trained assistants take over.

This information is provided for guidance only. It is recommended that persons associated with the installation of electrical equipment or repair of electrical installations obtain Australian Resuscitation Council approved training in current resuscitation methods.